



CHAPTER

Concentrate Tank

Buffers and recirculates the brine/reject stream, and doubles as the CIP reservoir

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16.1 Overview and function

The concentrate tank (tag prefix **TC**) buffers the brine/reject stream as it recirculates through the SonixED™ stack(s) (chapter 10): rejected ions accumulate in this stream over multiple passes, increasing its conductivity, until it is drained for disposal or routed onward for brine valorization (chapter 3). This tank also does double duty as the **circulation reservoir during CIP cleaning** — HydroVolta's documented CIP procedure (chapter 12) explicitly circulates acid and hypochlorite cleaning solution through this same tank, which is why the concentrate tank's minimum level is a precondition for starting a CIP cycle.

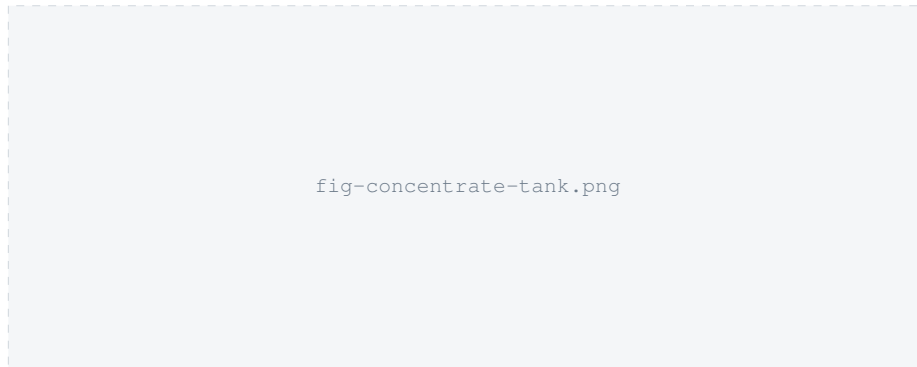


Figure 16.1. Suggested figure: photograph of the concentrate tank, labeled with its connections (from-stack, to-stack recirculation, drain/valorization outlet, CIP circulation port).

16.2 The CIP reservoir: water type, chemicals and dangers

No document identifies a separate, dedicated CIP tank — the concentrate tank above is the one and only vessel used for CIP, per HydroVolta's own CIP procedure (chapter 12). What is *in* the tank changes completely depending on which mode the unit is in, and confusing the two is a real hazard, so treat this section as a supplement to chapter 12, read from the tank's point of view.

- **During Production mode:** the tank holds **concentrate/brine** — rejected feedwater ions, increasingly concentrated over multiple passes (*Overview and function* above). This is process water, not a cleaning chemical.
- **During the CIP acid step:** the tank holds **pretreated feedwater dosed with HCl to pH 2.5** (chapter 12) — an acidic solution, corrosive at that pH.
- **Between CIP steps:** the tank is drained and refushed with **fresh feed water** for 15–20 minutes specifically to clear the acid before the next chemical goes in.
- **During the CIP hypochlorite step:** the tank holds **feedwater dosed with NaOCl to 2 ppm** (sodium hypochlorite, effective only near pH 7–7.5) — an oxidizing, corrosive solution.
- **During the NaOH variant of CIP** (electrolyte-compartment cleaning): the tank holds **NaOH dissolved only in pure (permeate-quality) water** — **never** feedwater or process water, per the warning in chapter 12 (mixing NaOH with the minerals present in feed/process water causes scaling).



DANGER

Never let the acid (HCl) and hypochlorite (NaOCl) steps mix in this tank. Mixing an acid with sodium hypochlorite releases chlorine gas, which is toxic even in small quantities. This is precisely why HydroVolta's documented procedure requires a full drain and a 15–20 minute fresh-water flush between the acid step and the hypochlorite step (chapter 12) — never shorten or skip that flush to save time, and never dose both chemicals into the tank at once. If chlorine odor is detected during or after a CIP cycle, stop dosing immediately, ventilate the area, and do not re-enter until it has cleared — see section 2.5 for first-aid measures for chemical/inhalation exposure.

Both the acid and hypochlorite solutions are corrosive at the concentrations used (see the Safety Data Sheet summary in section 2.5): treat the tank, its vent, and any sample point as a chemical-exposure risk during CIP, not just during dosing-pump refilling. Wear the PPE specified in section 2.3 whenever the tank is opened, sampled, or drained during or immediately after a CIP cycle.

Cleaning the tank itself: the documented procedure already *is* the tank-cleaning procedure — there is no additional physical scrub-down step on file. In order: drain the acid step, flush with fresh feed water 15–20 minutes; drain the hypochlorite step 5 minutes, then flush with fresh feed water 30 minutes (chapter 12). TODO: no document describes a physical inspection (scale/residue check on the tank walls, fittings, level sensor) as part of or after this cycle — until one exists, a visual check after the final flush (before returning to Production mode) is good practice.

16.3 Key specifications

As with the diluate tank, no finalized standard-product capacity/ material specification exists — see chapter 15 for the two real tank orders on file, which are candidates but not confirmed as to role. Some documented configurations use **two** concentrate buffer tanks in series (tagged **BTC1/BTC2**) rather than one.

- Level: TC_Level1 (main tank); BTC1/BTC2 level alarms exist in some configurations.
- Conductivity: monitored via the “Concentrate Sensor Island” (candidate: KROHNE SMARTPAT COND, chapter 13).
- Flow: concentrate-stream flow instrumentation confirmed in the register lists reviewed (chapter 13); a documented alarm exists for concentrate flow below **500 l/hr** (appendix A).

16.4 Operation

Concentrate recirculates through this tank during **Production** mode, increasing in conductivity pass over pass; during **CIP** mode (chapter 12), the same tank instead circulates cleaning chemical. Do not start a CIP cycle without first confirming the concentrate tank's level and that the CIP injection nozzle is submerged (chapter 12).

16.5 Maintenance

TODO: no tank cleaning/inspection schedule was found. Because this tank carries the concentrated reject stream (and periodically, cleaning acid/ hypochlorite), scaling and chemical residue are more likely here than in the diluate tank — inspect and descale per chapter 7 once an interval is confirmed, and always flush with fresh feed water after a CIP cycle per the procedure in chapter 12.

16.6 Troubleshooting

HL_Alarm_TC/LL_Alarm_TC cover the main tank (plus equivalent alarms for BTC1/BTC2 where fitted); a documented alarm fires below 500 l/hr concentrate flow. On any level or flow alarm, check the relevant sensor per section 13.4 before assuming a process fault. If a CIP cycle fails to reach its target pH, verify the concentrate tank's level and injection nozzle first, per chapter 12.

